1.1.3 Electricity

It is a kind of energy that one cannot see as like air. It's existence can only be known through the working of electrical appliances. Also its existence can be found using line tester and multimeter.



Initially, electricity was known as electric fluid.

1.2 TYPES OF ELECTRICITY

Generally there are two types of Electricity. They are:

1. Alternating Current (AC)
2. Direct Current (DC)

1.2.1 Alternating Current

The direction of current varies with time is called as alternating current. It has two terminals viz., phase and Neutral. If the phase terminal acts as '+VE', then neutral terminal acts as '-VE' terminal. If phase terminal acts as '-VE', then neutral terminal acts as '+VE' terminal.

If you test the phase terminal with a line tester, you may see the tester shows light indication. Neutral shows no indication. Neutral is a common terminal used as a return path current flowing from phase to neutral through the connected device. It is neither '+VE' nor –VE.

1.2.2.1 Voltage and Frequency of Household AC Supply

In our country, the voltage and frequency of single phase supply is maintained as 220V/50Hz. In Foreign Countries, particularly in U.S.A, it is maintained as 110V/60Hz AC. Figure 1.1 shows the waveform of the alternating current.

1.2.2.2 Waveform

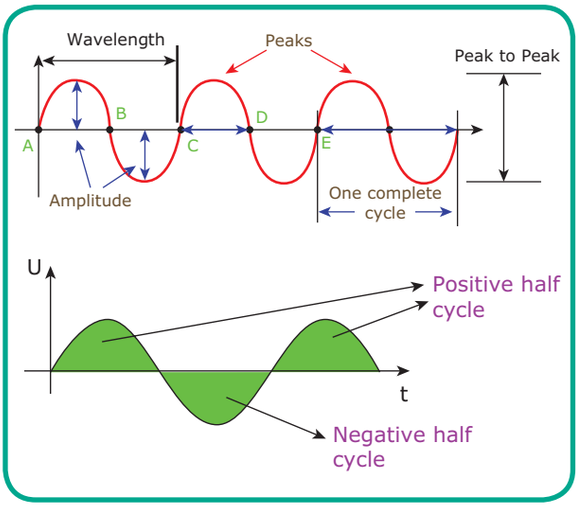


Figure 1.1 AC sine waveform

1.2.2.3 Amplitude

Amplitude is the distance from the X-axis to the vertex or peak of the any one half cycle of the AC waveform. Its unit is volts.

1.2.2.4 Polarities

The top half cycle is positive half cycle and the bottom half cycle is negative half cycle. Both half cycles combined together to form the one complete full cycle.

1.2.2.5 Wavelength

The length from starting point of the wave to the finishing point, which includes one positive half cycle and one negative half cycle, is called wavelength. Its unit is meter.

1.2.2.6 Peak to Peak Value

The distance between positive peak and negative peak is referred as peak to peak value. Its unit is volt.

1.2.2.7 Angles

The starting point of the wave is 0°, +VE peak is 90°, the end of positive half cycle is 180°, peak of negative half cycle is 270°

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